



Trimble® Earthworks Towed Scraper

For SITECH Only

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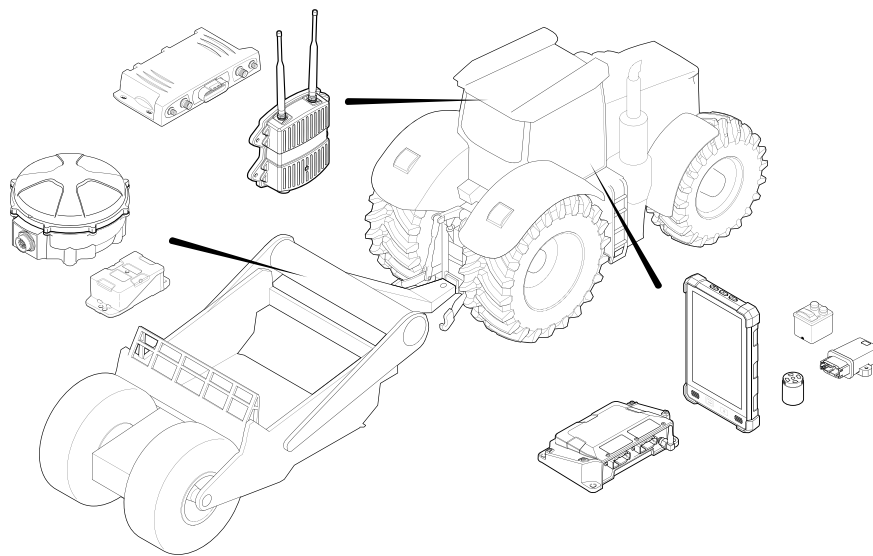
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Where the components go on the machines



Towed scraper hardware install checklists

In this guide:

- ▶ [Checklist—Pre-installation](#)

2.1 Checklist—Pre-installation

Use the following checks to identify problems with a machine before an installation. Problems that are identified by this procedure must be corrected before commissioning is begun. Poor machine condition can affect the performance of the automatic blade guidance.

- ☐ Operate the machine and observe the performance of the hydraulic system, particularly the responsiveness of the hydraulic system to control movements. Any hydraulic system issues will remain after installation, and may be worsened if an Autos system is installed.
- ☐ Check the cutting edge:
 - ─ Even vertical wear over the entire length of the blade. 0 ± 5 mm horizontal deviation.
- ☐ Check the hydraulic rams and joints: place the bowl on flat ground. Activate the rams to raise the blade and ensure there is limited play in the joints.

Tools and other supplies

In this guide:

- ▶ Replacement serial number plates
- ▶ Extra extension cables
- ▶ Tools

3.1 Replacement serial number plates

If you need to install a system component over a serial number plate on the machine, you can order a replacement serial number plate from your dealer. You can then install the replacement serial number plate in another location on the machine. To maintain your machine's identification, you must obtain your dealer's approval to relocate the serial number plate before removing the existing serial number plate.

3.2 Extra extension cables

Always take a extra extension cables of varying lengths to an installation. The extension cables included in the kit may not be long enough for each machine's specific circumstances.

3.3 Tools

Make sure that you have the following tools and supplies available:

- Arc Welder / Welding Rods
- Volt/Ohm Meter
- Mechanical and Electrical hand Tools
- Clean Container for Hydraulic Oil
- Clean Rags

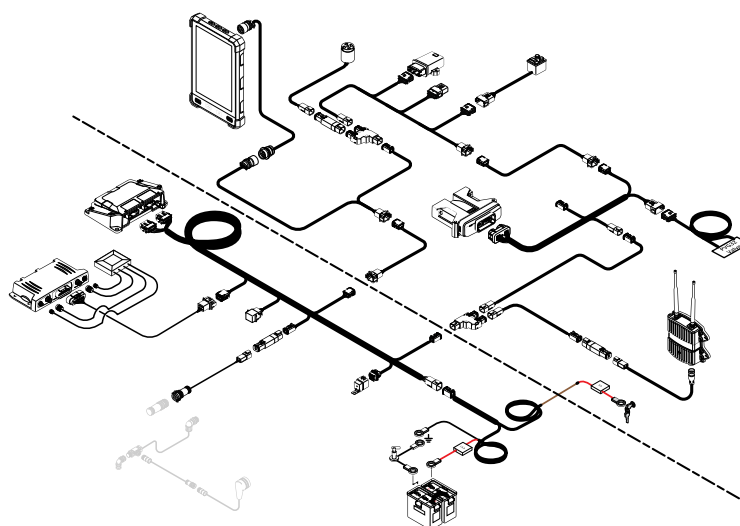
Tractor cabling layout guide

In this guide:

- ▶ Overview
- ▶ Power cable routing
- ▶ Component mounting locations

4.1 Overview

This guide describes how to layout the component cables in the cab on any machine that will tow a towed scraper.



Some users may replace the TD5x0 with a supported third party tablet.

4.1.1 Kit list

Use the following kits to install the towing machine cab components:

Kit number	Description
Required	
160021-500	Base, Towed Scraper, Generic, V3.1
Optional	
160021-501	Valve Module Add-on, Towed Scraper, Generic, V3.1
160041-10	Kit - SNM941 roof mounting
112057-00	Kit - SNM941 4-in-1 antenna with mounting bracket
Supported third party tablet (optional)	You may swap the TD5x0 for a third party tablet. Talk to your dealer about which tablets are supported.

4.1.2 Cable routing best practice

When selecting a route for cables and harnessing consider the following:

- Exposure to hazards, including:
 - mechanical damage by the operator
 - environmental hazards
 - heat
 - corrosive material
- Existing grommated entry points. When taking the cable outside the cab, use grommated holes to minimize dust and moisture into the cab.
- Moving parts
- Sharp edges
- Easy access for replacing the cable, if needed in the future

4.2 Power cable routing

NOTE –Turn off the master disconnect switch before connecting to the power source on the machine.

4.2.1 Power connections, components, and cables

Part number	Items	Description
150603-02	1	EC520 long platform harness
150411-045	1	Power cable
109333-12	1	Relay* (12V)
109333-24	1	Relay* (24V)

* The kit includes both a 12V and a 24V relays machine. Use the relay that is appropriate for the machine voltage.

To connect:

1. Route the power cable, between the EC520 long platform harness and the battery.
2. Connect the relay, as shown in the connection diagram.

4.3 Component mounting locations

This section shows possible installation locations of cab components. Read the install section for each component for hints and tips for installing that component, then choose the best location for the component on your machine.

4.3.1 System harness

Consider and how the components in the following sections connect to the system harness. Drape the system harness inside the cable to decide where to route the harness so you can attach the components.

4.3.2 Display and remote switch

Discuss with the operator where to install the components the operator will interact with:

- Display - attach the Ram ratchet ball-clamp to the ball and the display and trial various positions
- Remote switch - attach to either the joystick or a rail

Connect the display and remote switch cables to the system harness.

4.3.3 EC520 control module

The EC520 control module works as a computer only and does provide sensor readings. You can install the EC520 in any location that:

- Will not interfere with the operator
- Does not transfer heat to the EC520 or the harness
- Gives enough space to route and bend cables from the EC520 into the cab, ensuring that you only bend cables within the tolerances specified in the instructions for the machine-specific component kits

Suggested locations for the EC520:

- Under one of the plastic panels
- Behind the seat
- Under the cab

Connect the EC520 cable to the system harness.

4.3.4 AA510, CAN interface module and valve module

Connect the AA510, CAN interface module and valve module cables to the system harness and consider where you can install these components.

4.3.5 SNM94x Connected Site Gateway

NOTE –Do not weld any components at this stage. Carefully mark the locations for the weld-mounted components, and then route the cables to make sure they will reach the components when they are installed.

NOTE –Follow the connection diagram to check the cable reaches the proposed location of the bracket. Do not cut the cable. Use extension cables where needed.

The primary consideration is for operator and technician safety:

- Rotate the SNM94x Connected Site Gateway so that the connector is along the most convenient side and you can access the SIM card. If you rotate the bracket, move the stowage blank appropriately.

For optimum performance:

- Connect any antennae above the roof line with an uninterrupted view.
- The lengths of the cable between antennae and the SNM94x Connected Site Gateway are fixed. Do not use extension cables as extension cables reduce performance. Do not shorten the antenna cables as cutting cables violates some regulations.
- To prevent signal interference the antennae must be at least 200 mm away from any other transmitting antenna, 100 mm away from any receiving antenna, and higher than surrounding metal objects.
- If using an SNR radio, the antenna must be 100 mm away from either of the SNM94x antennae.

To avoid damage:

- To reduce damage to the SNM94x from shock and vibration, position the bracket on a solid part of the cab

4.3.6 Quick disconnect

Consider where to install the quick disconnect bracket in a location that is:

- Free from interference from moving parts
- Easily accessible for connecting and disconnecting the towed scraper harness

Connect the quick disconnect cable to the system harness. Route the cable between the system harness and the quick disconnect bracket location. Use extension cables if needed.

Tractor install guide

In this guide:

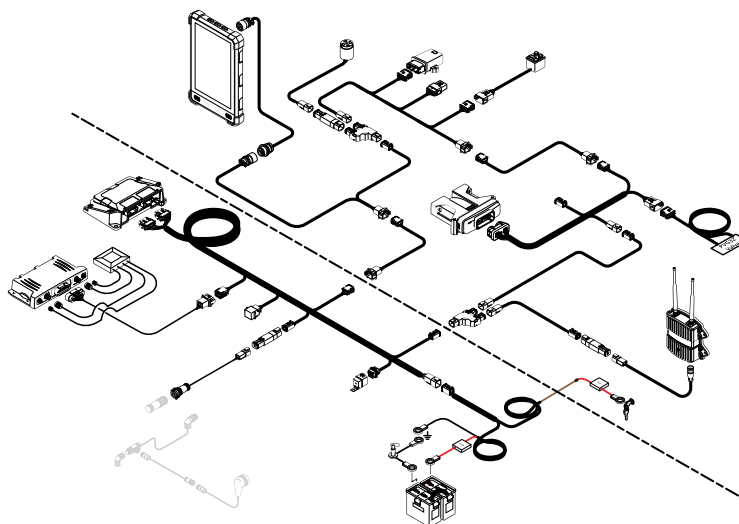
- ▶ Overview
- ▶ EC520 control module install
- ▶ TD540 display install
- ▶ CI5xx CAN interface module install
- ▶ Installing the AA510 audible alarm
- ▶ Remote switch install
- ▶ Valve module install

5.1 Overview

This guide describes how to install the components in and around the cab on any tractor that will tow a towed scraper.

For more detailed hardware and cabling installation instructions, refer to the machine-specific instructions that accompany the component kits.

5.1.1 Cab components, cables and connections



Some users may replace the TD5x0 with a supported third party tablet.

5.1.2 Kit list

Use the following kits to install the towing machine cab components:

Kit number	Description
Required	
160021-500	Base, Towed Scraper, Generic, V3.1
Optional	
160021-501	Valve Module Add-on, Towed Scraper, Generic, V3.1
160041-10	Kit - SNM941 roof mounting
112057-00	Kit - SNM941 4-in-1 antenna with mounting bracket
Supported third party tablet (optional)	You may swap the TD5x0 for a third party tablet. Talk to your dealer about which tablets are supported.

5.1.3 Prerequisites

Check before starting:

- Both the 12V and 24V relays are included in the kit. If the machine is a
 - 12V machine -use the 12V relay, P/N 109333-12
 - 24V machine, use the of the 24V relay, P/N 109333-24.
- Do not remove the backing film or the sorbothane layer on the EC520. The film protects the sorbothane layer.
- If you are using a supported third party tablet as the display, there are some important differences in the install procedure. Refer to [Chapter 9, Supported third party tablet installation](#) for more information.

5.1.4 Connecting to machine power

NOTE –If you connect the system's ground (negative) connection to the negative terminal of the battery, and attempt to start the machine with the master disconnect switch open (or if the master disconnect switch or battery ground cable is defective), then the starter current will flow from a grounded component on the chassis to the battery terminal and damage the system. Make sure that you connect the system's ground connection to the machine frame.

TIP – While you are connecting the power, place a "Machine under service, do not turn on" tag on the battery isolation switch.

Constant power and key switched power connections are made on the main run relay for the machine, using the power cable.

The ground (Gnd) connection is made at the main chassis connection for the machine, located above the batteries.

5.2 Component mounting locations

This section shows possible locations of components in and around the cab. Read the install section for each component for hints and tips for installing that component, then choose the best location for the component on your machine.

5.2.1 EC520 control module

Requirements:

- Attach the EC520 bracket to the machine.
- Do not cut the cables. Use extension cables where needed.
- You cannot drill into, or weld on, any part of the machine's Roll Over Protection Structure (ROPS) as this could weaken its structural integrity.

NOTE –Many cabs are now an integral part of the machine's ROPS, and are certified as such. Drilling into, or welding on, these cabs may invalidate their ROPS certification.

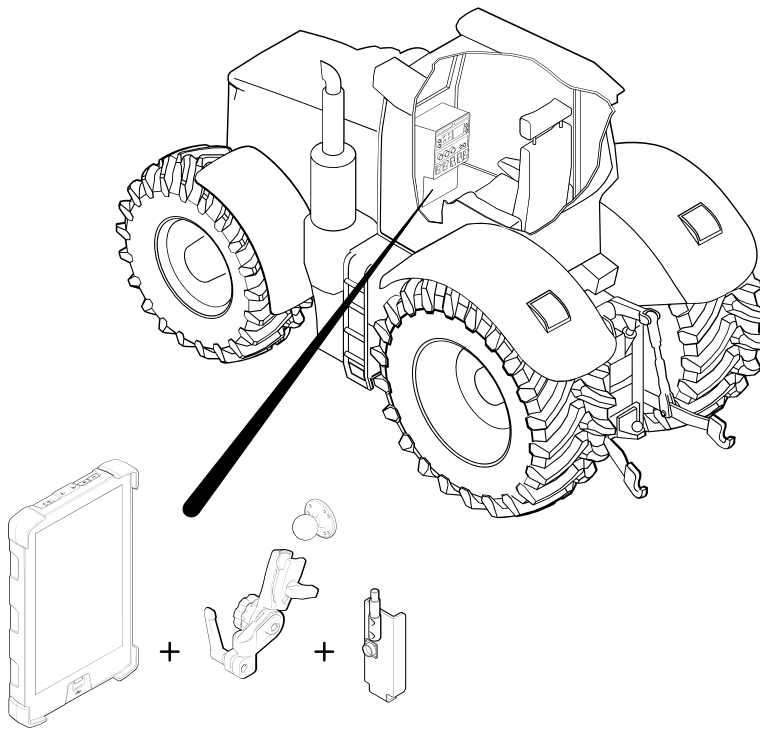
To avoid damage, consider:

- A location and orientation that prevents water collecting in the EC520 connectors.
- A location or surface that does not transfer heat to the EC520 or the harness.
- Space to route and bend cables from the EC520 into the cab, ensuring that you only bend cables within the tolerances specified in the instructions for the machine-specific component kits.

Suggested locations for the EC520:

- Directly under the cab, if such a surface exists. This may be above a plate protecting the under side of the cab.
- In the engine compartment.
- Behind the cab.
- In the cab, under one of the plastic panels.

5.2.2 TD540 display



To select a route for the display cable consider the following:

- Exposure to hazards, including:
 - mechanical damage by the operator
 - environmental hazards
 - heat
 - corrosive material
 - weather
- Existing grommeted entry points. When taking the cable outside the cab, use grommeted holes to minimize dust and moisture into the cab.
- Maintaining free space between structures holding the display, or the display itself. Maintain free space between the display and windows that open and close, controls, and the view of the work area.
- Access to the remote switch cluster, joystick, and existing machine harness
- Access to the AA510
- Moving parts
- Sharp edges

Try various positions:

- Attach the Ram ratchet ball-clamp to the ball and the display and trial various positions.

For more information about attaching the mount, see [5.4 TD540 display install](#).

5.2.3 SNM94x Connected Site Gateway

NOTE –Do not weld any components at this stage. Carefully mark the locations for the weld-mounted components, and then route the cables to make sure they will reach the components when they are installed.

NOTE –Follow the connection diagram to check the cable reaches the proposed location of the bracket. Do not cut the cable. Use extension cables where needed.

If using an SNM94x Connected Site Gateway, the primary consideration is for operator and technician safety:

- Rotate the SNM94x Connected Site Gateway so that the connector is along the most convenient side and you can access the SIM card. If you rotate the bracket, move the stowage blank appropriately.

For optimum performance:

- Connect any antennae above the roof line with an uninterrupted view.
- The lengths of the cable between antennae and the SNM94x Connected Site Gateway are fixed. Do not use extension cables as extension cables reduce performance. Do not shorten the antenna cables as cutting cables violates some regulations.
- To prevent signal interference the antennae must be at least 200 mm away from any other transmitting antenna, 100 mm away from any receiving antenna, and higher than surrounding metal objects.
- If using an SNR radio, the antenna must be 100 mm away from either of the SNM94x antennae.

To avoid damage:

- To reduce damage to the SNM94x from shock and vibration, position the bracket on a solid part of the cab.

Attach SNM94x Connected Site Gateway to bracket

For specific installation instructions, refer to the instructions provided with the radio bracket kit (P/N 109536).

Validate

To check you have installed the SNM94x Connected Site Gateway and Wi-Fi antenna correctly, turn on the system.

- After a couple of minutes the SNM94x Connected Site Gateway will appear in the list of devices in the Web Interface.
- Check that the Web Interface's Monitor > Onboard Devices screen shows that the connection to the gateway is good.

5.2.4 Brackets for Wi-Fi antenna and cellular + GNSS antenna

The SNM941 uses a single integrated Wi-Fi antenna and cellular network antenna + L1 GNSS patch antenna.

5.2.5 SNR radio

If using an SNM94x, you can mount the SNR radio on top of the gateway module bracket, if you are using a gateway module.

If attaching the SNR radio to any other part of the machine, make sure the part is well braced as the extra weight of the radio and bracket can bend some machine parts.

Consider the following:

- Check that the radio cable reaches from the EC520 long platform harness to the radio bracket on the cab.
- The location of the SNR rover radio connector and the route for the SNR radio cable.

5.2.6 Valve module and CI5xx CAN interface module

Mount the valve module and the CI5xx CAN interface module inside the cab using the following mounting devices:

- Valve module - P/N 92609 Angle bracket
- CI5xx CAN interface module - the velcro strap included in the kit

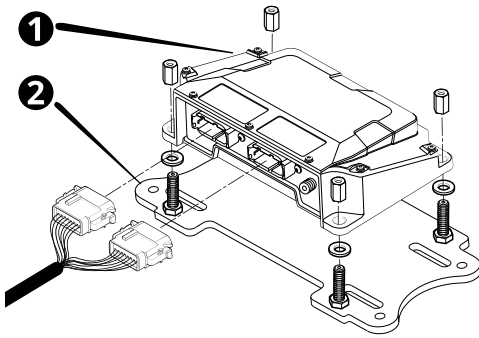
5.2.7 Installing SNM941 antenna

Install the SNM941 antenna to send and receive data to the office including, firmware updates, diagnostic files, position information and design files.

Refer to the *SNM941 Connected Site Gateway Installation Guide* for a detailed description of the modem installation.

5.3 EC520 control module install

This section describes how to install the EC520 control module.



Item	Part number	Description
❶	100435-10	EC520
❷	201910-10	Bracket

5.3.1 Connections, components, and cables

Part number	Items	Description
150603-02	1	EC520 long platform harness
150411-045	1	Power cable

5.3.2 Install the EC520 control module

1. Decide on a installation location on the towing machine.
2. Place the bracket on the installation surface.
3. Using the bracket as a template, drill holes at the outside edges of the wide slots.
You cannot drill into, or weld on, any part of the machine's Roll Over Protection Structure (ROPS) as this could weaken its structural integrity.

NOTE –Many cabs are now an integral part of the machine's ROPS, and are certified as such. Drilling into, or welding on, these cabs may invalidate their ROPS certification.

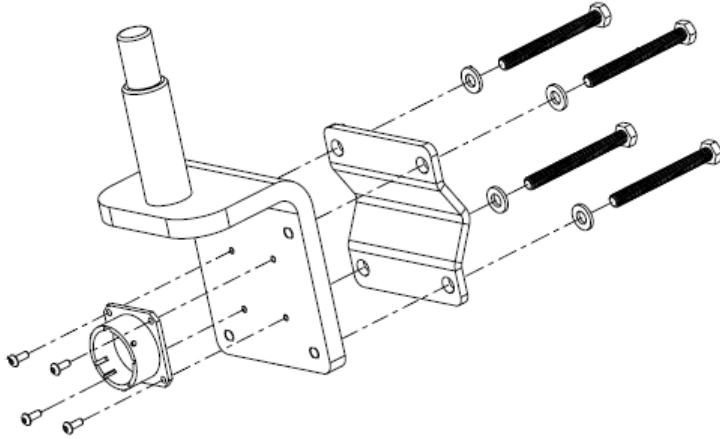
4. Clamp the bracket to the machine.
5. Attach the EC520 to the bracket using the coupling nuts.
6. Connect the EC520 to the cabling as shown in the connection diagram. See [Cab components, cables and connections](#).

5.4 TD540 display install

This section describes installing the TD540 display using a ratchet clamp and ball mount (P/N 58440-10). There are two options for mounting the ratchet clamp and ball mount to the inside of the cab:

- Attach a handrail bracket mount (P/N 101808)
- Attach a magnetic display mount (P/N 110262-00)

5.4.1 Attach a handrail bracket mount



Part number	Description
101808	Handrail bracket mount
58440-10	Ram ratchet ball-clamp

To install:

1. Decide on the location of the display, such as a bar or handrail in the cab that is within easy reach for the operator.
2. Attach the handrail bracket to the desired location.
3. Attach the Ram ratchet ball-clamp.
4. Attach the display.
5. Attach the display cable and any extension cables required to reach the EC520 long platform harness, P/N 150603-02.

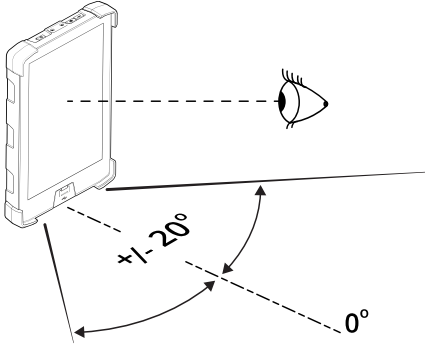
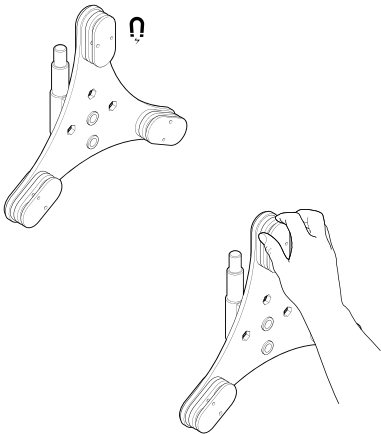
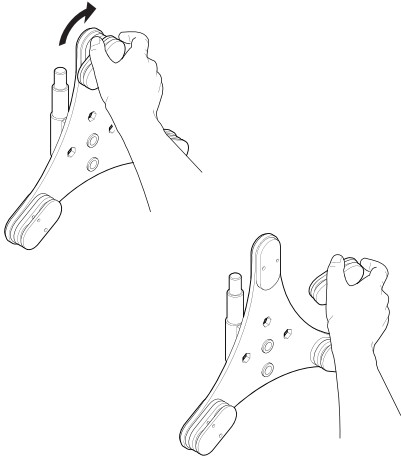
5.4.2 Attach magnetic display mount

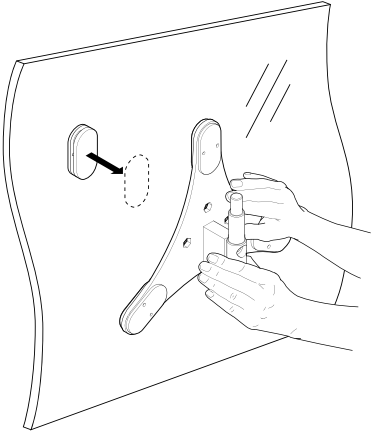
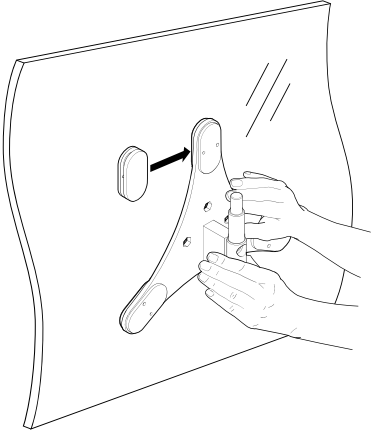
NOTE –It is easier to perform this task with two people.

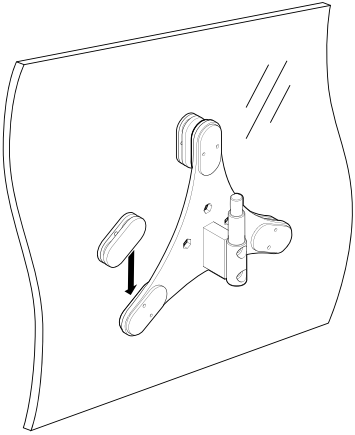
NOTE –Mount magnetic display mounts to flat glass only.

WARNING –The magnets on the magnetic display mount are strong. If parts of your hands get between two magnets when you move them, the magnets may suddenly move and pinch or clamp the skin. When the magnets are close together, position your hands so they are not between the two magnets.

WARNING –If you have a pacemaker and handle the magnetic display mount, the magnet can affect the function of the pacemaker and cause physical harm. Remain at least 30 cm (1 foot) from the magnetic display mount if you have a pacemaker.

Steps	Description
	Decide on the location of the display.
	Hold the magnetic display mount and carefully twist the magnet and tri-arm in opposite directions.
	When the force between the magnets weakens, pull the tri-arm and magnet apart. Remove the padding and keep the magnets away from each other.

Steps	Description
Practice connecting and disconnecting.	Using two people and the padding, practice placing the tri-arm and magnet together. Keep the magnets close but not opposite each other with the padding in between, then rotate and slide the magnet until the magnets connect.
	On the machine, place the tri-arm magnetic assembly against the inside cab window in the required position. Check the post is aligned correctly and that the cables, Ram ratchet ball-clamp, and display are in the correct position. Remove the Ram ratchet ball-clamp and display before continuing. Hold the interior tri-arm firmly in the required position and make sure all three magnets touch the glass.
	On the outside of the glass, attach the first magnet. • Once the magnets connect they are very hard to disconnect. • Keep the magnet surfaces away from dirt or other metals. • Start with the outside magnet away from the inside magnet, then slide toward each other.

Steps	Description
	Repeat for all three magnets.

To finish:

1. Attach the Ram ratchet ball-clamp.
2. Attach the display.
3. Attach the display cable and any extension cables required to reach the EC520 long platform harness, P/N 150603-02.

5.4.3 Validate

1. Connect the display cable to the TD5x0 display. If using a supported third party tablet as the display, talk to your dealer.
2. Turn on the ignition. The display starts automatically.
3. Tap the Earthworks icon. Earthworks opens.
4. If the log in screen appears, then the display is connected to the EC520.

Some of the problems that you might see include:

Problem	Description	Resolution
The display does not start.	There is no power to the display.	Check connectors between the display and the EC520. Use the electrical schematic to check the pin numbers from the battery.

Problem	Description	Resolution
Using my own display, and not the TD5x0, I cannot connect to the EC520.	The first time you run Earthworks, there is no Wi-Fi connection with the EC520.	Connect a laptop, or similar, to the EC520 long platform harness "Display, Eth1 + CAN2" connector. On the laptop, open the Web Interface, log in, and set up the Wi-Fi access point on the EC520.
On startup, I received a warning to upgrade the display operating system.	An important operating system update for the display is available.	Contact your system dealer to schedule the update.

5.5 CI5xx CAN interface module install

The CI5xx is a Controller Area Network (CAN) adapter that converts the remote switch increment and decrement inputs to CAN. Only the CI520 CAN interface is supported.

5.5.1 Attach CI520

- Dual locking strip
- Cable ties
- CI520 CAN Interface Module, P/N 132739-10
- Cable, 150820-01 Rev B

To install:

1. Connect the CI520 to the cabling as shown in the connection diagram.
2. Place the CI520 inside, or under, the cab. It is often behind the seat.
3. Either use zip ties or use the Velcro supplied in the kit to secure the CI520, if required.

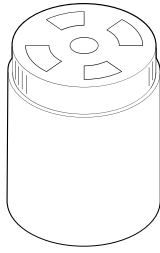
5.5.2 Validate CI520

After connecting the display and running Earthworks, select System Status > Remote Switches. The remote switch status should be Active and the model should be CI520.

5.6 Installing the AA510 audible alarm

The AA510 alerts the operator to whether they are on grade, above grade or below grade. The AA510 attaches to the system via a CAN bus.

5.6.1 Attach AA510



- Rubberized p-clamp P/N 97587-45
- Cable ties

To install:

1. Decide where to locate the AA510.
2. Connect the AA510 to the EC520 long platform harness, as shown in the connection diagram.
3. Use P-clamp or cable ties to attach the AA510 to a fixed surface or component.

5.6.2 Validate

After connecting the display and running Earthworks, test the AA510 using the System Settings > Sound option in Earthworks.

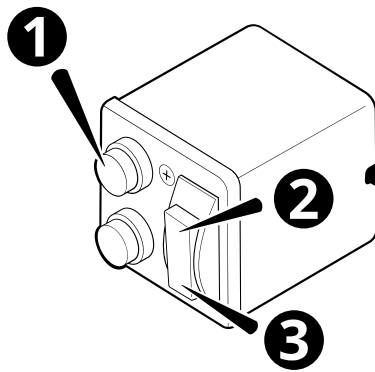
Some of the problems that you might see include:

Problem	Description	Resolution
AA510 too quiet or too loud.	The AA510 might seem loud in one machine and not loud enough in another machine because of where you locate the AA510 and the machine noise.	In the Operator App, open Sound Settings and edit the volume of the AA510 and the type of event that sets off the AA510.

5.7 Remote switch install

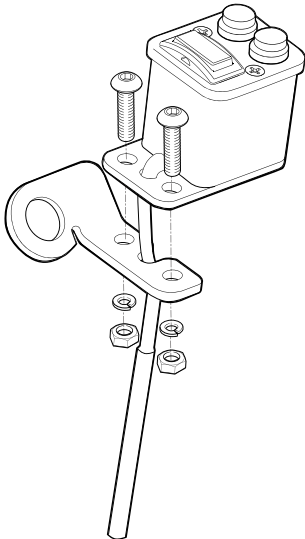
The towed scraper system can have one vertical offset remote switch for:

- Activating the automatic controls
- Incrementing or decrementing the elevation offset or target slope



- | | | | |
|---|------------------------------------|---|------------------------------------|
| ❶ | Auto switch. | ❷ | Increment elevation offset switch. |
| ❸ | Decrement elevation offset switch. | | |

5.7.1 Remote switch assembly



Part number	Description
150420-025	Remote switch assembly

5.7.2 Install the switch harness

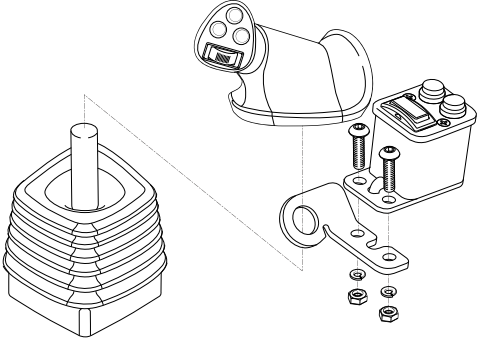
Select a route for the cable by considering:

- Exposure to hazards, including:
 - mechanical damage by the operator
 - environmental hazards

- heat
- corrosive material
- Access to the CI5xx cable
- Moving parts
- Sharp edges

Discuss with the machine operator where in the cab to attach the remote switch.

Install and connect the remote switch

Illustration	Steps
	Vertical offset remote switch install

To install:

1. Attach the supplied mounting tab to the base of the remote switch.
- NOTE** –If your joystick cannot be detached from the control assembly, replace the mounting tab in the kit with the joystick handle mounting tab.
2. Attach the mounting tab to the joystick.
 3. Route and secure the remote switch and connect the cable to the cab harness.

NOTE –If it is not practical to bolt the remote switch somewhere in the cab, use double sided tape to attach the remote switch to a location within easy reach for the operator.

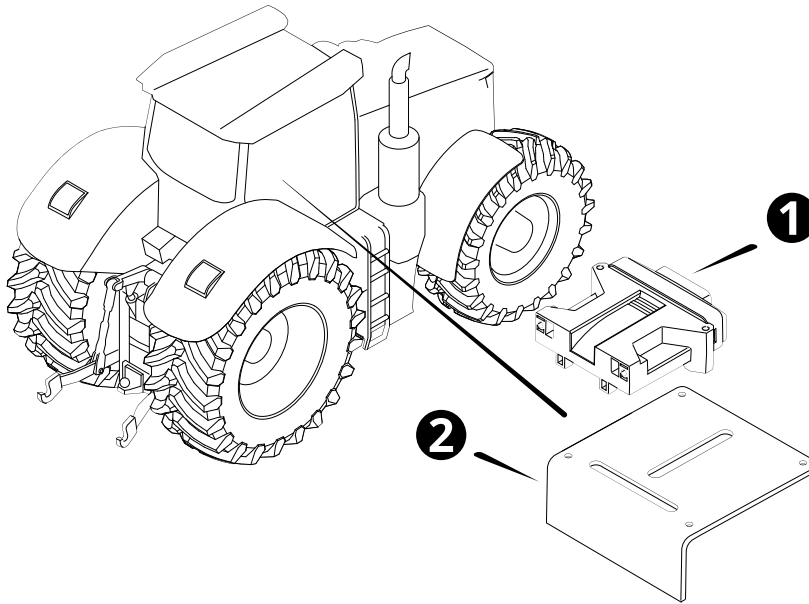
5.7.3 Validate

Some of the problems that you might see include:

Problem	Description	Resolution
Remote switch does not work.	The switch is not connected.	Check the switch connects to the system by opening the Web Interface and checking Monitor > Onboard Devices. Check the connections.

5.8 Valve module install

The valve module provides precise control of the bowl's lift function by sending output signals to the machine's electro-hydraulic valves or the hydraulic kit installed on the bowl as part of this system.



Item	Part number	Description
❶	92609	Module angle mounting bracket
❷	100441-00	VM510

5.8.1 Connections, components, and cables

Part number	Items	Description
150906-01	1	Valve module harness
86930-05	1	Valve cable
150858-01	2	Splitter, 4 Circuit
150473-020	1	Cable extension

5.8.2 Install the valve module

1. Attach the module mounting bracket to the towing machine. Either bolt or weld the bracket to the machine.
2. Attach the valve module to the module mounting bracket.
3. Connect the cabling to and from the valve module according to the connection diagram for towed scraper, available on [Partners](#).

5.8.3 Validate

After connecting the display and running Earthworks, use the Web Interface's Configure > Valve Test screen to test that the valves are driving the lift rams.

Tractor main harness and quick disconnect install guide

In this guide:

- ▶ Overview of the install
- ▶ Connections, components, and cables
- ▶ Platform harness routing
- ▶ Quick disconnect install

6.1 Overview of the install

This guide describes the harness cables and connectors for a generic towed scraper system, and applies to a new machine or when replacing components on a machine currently using grade control.

6.2 Connections, components, and cables

Part number	Items	Description
150603-02	1	EC520 long platform harness
150473-020	1	Cable extension, 2m, 4 pin
150413-010	1	Cable extension, 1m, 4 pin
150857-01	1	Inline terminator
132467-050	1	Harness extension, 5m
64101	1	Bracket, quick disconnect
	4	Dust caps

6.3 Platform harness routing

One of the easiest ways to see how the system fits best on your machine configuration is to drape the cables in possible locations, adding extension cables where needed and protecting any places where the cable will rub against edges.

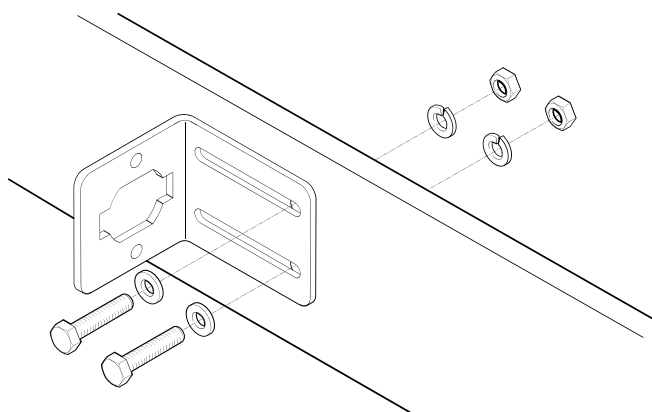
Using a combination of splitters (P/N 150858-01), inline terminators (P/N 150857-01), and the cables listed in [Connections, components, and cables](#), connect the system hardware to the EC520 long platform harness.

NOTE – Depending on the size of the machine configuration, more or fewer extension cables (P/N 150416-0x0, where x indicates the length of the cable in meters) may be required. The Towed Scraper Base Kit provides one 3 m and one 2 m extension cables.

6.4 Quick disconnect install

The quick disconnect is the connector between harness on the tractor and on the bowl. This connector will disconnect if the towed scraper is unhitched from the tractor but the harness is left connected. Consider this disconnect feature when deciding where to install the quick disconnect bracket.

The following figure shows a quick disconnect bracket. Bolt or weld the bracket to the back of the tractor.



6.4.1 Quick disconnects connections, components, and cables

Part number	Items	Description
64101	1	Quick disconnect bracket
PN 2117-1730	1	Dust caps

6.4.2 Requirements

Install the quick disconnect bracket (P/N 64101)

- on the rear of the tractor
- on the same side that the mast is installed on the towed scraper

Make sure there is enough space between the bracket and other fittings to provide access to the sensor cables.

6.4.3 Install and connect quick disconnects

1. Mount the quick disconnect bracket to a suitable location on the rear of the tractor.
2. Mount the quick disconnects into the bracket, aligning the spring clips with the openings at the top and bottom of the cutouts.
3. Insert the retainer clamp that holds the connectors in place.

NOTE –Insert the retainer clamp so that its thread insert faces the back of the bracket.

4. Secure the dust caps to the front of the bracket and clamp the connectors in place.
5. Connect the left/right splitter cable to the harness and route the connector to the quick disconnects.

Towed scraper install guide

In this guide:

- ▶ Overview of the install
- ▶ Connections, components, and cables
- ▶ Install the mast mount
- ▶ Install the sensor bracket

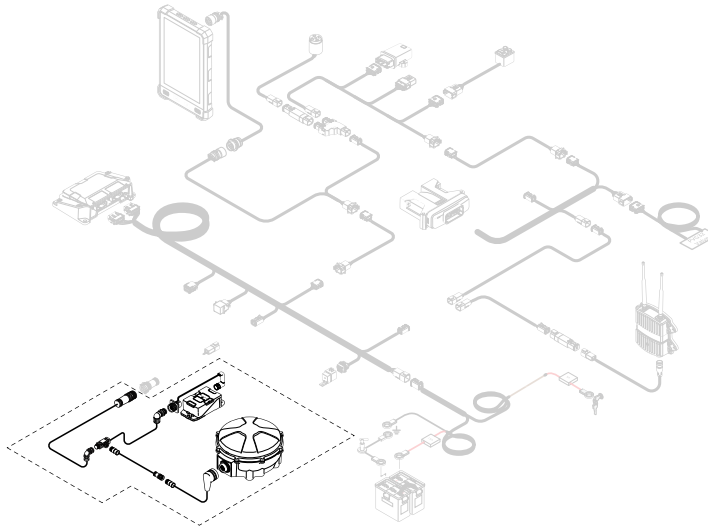
7.1 Overview of the install

This guide describes the harness cables, connectors and components for a generic towed scraper system. This information applies to a new machine or when replacing components on a machine currently using grade control.

Before you start to install the components:

- Position the machine on a flat and level surface with the cutting edge resting on blocks positioned so that the top of the cutting edge is level.
- Review this entire chapter. This will ensure that you understand the full installation workflow. It also lets you consolidate welding and fitting tasks, which may require external contractors, with other subsystem installations, such as the cab and blade installations.

7.2 Connections, components, and cables



Part number	Items	Description
78003-02	1	Kit – mast mount
160005-503	1	Kit – sensor bracket
81069-00	1	fixed mast
150603-02	1	EC520 long platform harness
150473-020	1	Cable extension, 2m, 4 pin
150418-030	1	Cable extension, 3m, 8 pin

7.3 Install the mast mount

These instructions apply to installing the mast mount to either the cross tube or the side of the bowl. The mast mount, which the IMU is also attached to, is the only hardware installed on the towed scraper.

NOTE –Installing the mast mount on a factory mast mount structure that reaches out over the cutting edge may affect guidance accuracy and is unsupported in this configuration.

7.3.1 Identify the mast mount install location

Decide where to install the mast mount. The location must:

- be a rigid surface that remains in a consistent position relative to the cutting edge at all times
- keep the mast vertical within +/- 5° with articulation when loading and dumping

The two locations on the bowl to consider for attaching the mast mount to are on the:

- cross tube (recommended install location)
- side of the bowl

NOTE –Masts mounted to the side of the bowl may rotate close to 90° during dumping and guidance may be lost.

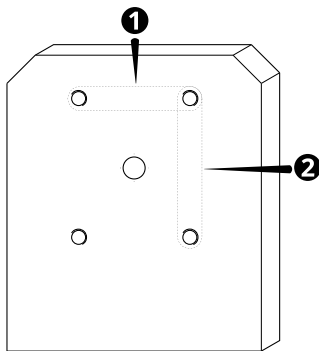
7.3.2 Position the mast mount

Position the cutting edge in its normal operating position and then check that the mast mount is:

- level in all directions (forward and back, left and right). Adjust if necessary.
- parallel to the cutting edge.
- in a position that will keep the mast and receiver clear of the apron.

The mounting plate has five pre-drilled holes. Position the:

- 100 mm (3.94 inches) spaced holes (❶) parallel to the cutting edge.
- 125 mm (4.92 inches) spaced holes (❷) parallel to the machine centerline.



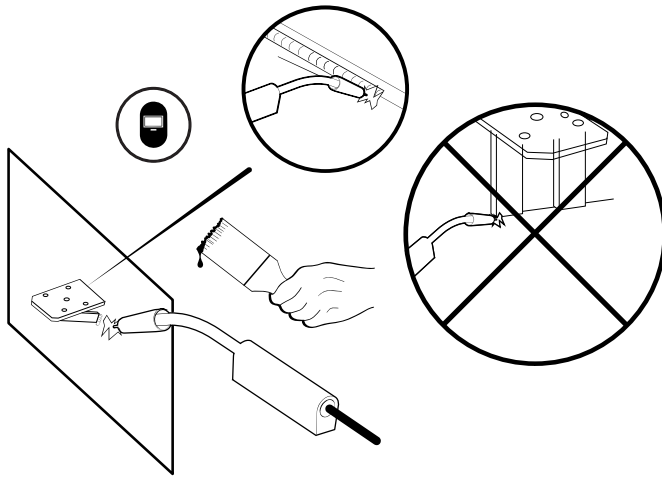
7.3.3 Weld the mast mounts

WARNING –Disconnect the negative battery terminal before commencing any welding and ensure that power is removed from the grade control system by disconnecting the power input cable or cables. Failure to follow this warning may result in electrical damage to system components.

TIP – Tack weld and check clearance to masts on the machine before welding completely. Raise the bowl to full height and run through full lower motion to ensure the masts and receiver installed on top of the mast will not come into contact with parts of the machine body or lift rams, lights, etc.

1. Tack weld the mounting plates to the mounting location.
2. Tack weld the fabricated gussets to an even surface.
3. Position the bowl in the normal operating position. You may need to sight the bowl on blocks. Check, and adjust as needed, that the mounting plate is :

- a. level in all directions (forward and back, left and right)
 - b. allows you to connect and disconnect the cables
4. When everything is correct, weld the mount around every edge to make it as rigid as possible.
5. Once the welding is completed, paint the mounting bracket and weld area where machine paint was removed.



7.4 Install the sensor bracket

Install the sensor bracket (P/N 160005-503) on top of the mast mount so that the sensor faces towards the tractor.

NOTE –Install the sensor bracket so the sensor faces towards the front of the bowl.

Towed scraper GNSS install guide

In this guide:

- ▶ Overview of the install
- ▶ GNSS connections, components, and cables
- ▶ Validate the GNSS system install

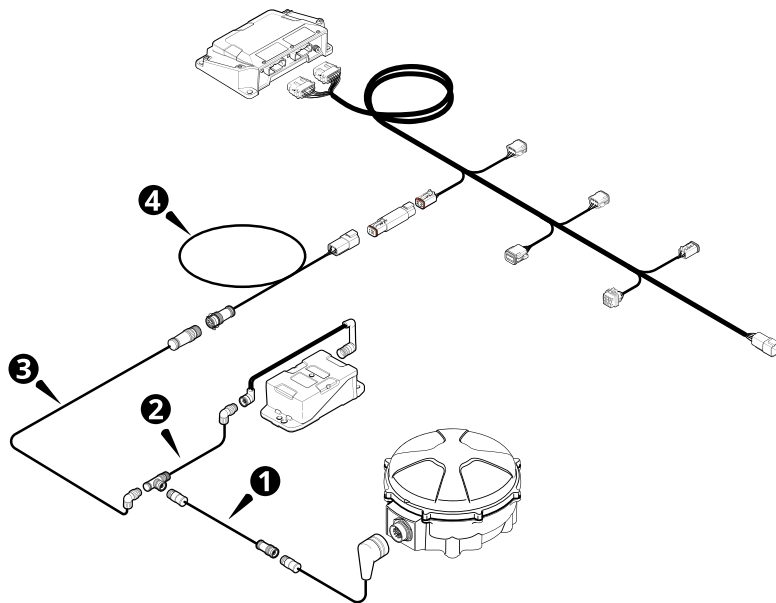
8.1 Overview of the install

This guide supports the addition of the hardware components of a GNSS positioning subsystem to a base Earthworks towed scraper system.

Before starting a GNSS subsystem install:

- Complete the installation of the cab components. In particular, if you are installing the optional radio then you need to install the SNM941 Gateway stacked radio bracket first.
- Review this entire guide. This will ensure that you understand the full installation workflow. It also lets you consolidate welding and fitting tasks, which may require external contractors, with other subsystem installations, such as the cab and front linkage installations.

8.2 GNSS connections, components, and cables



Item	Part number	Description
❶	0792-4410-060	Extender cable
❷	54485-06	GNSS T-cable
❸	150413-0xx	If required, GNSS Smart Antenna cables.

8.2.1 Components and cables from base kit (P/N 160021-500)

Part number	Provided	Description
0792-4410-060	1	Extender cable (❶)
150820-01	1	Splitter
150857-01	3	Terminator
54485-06	1	GNSS T-cable (❷)

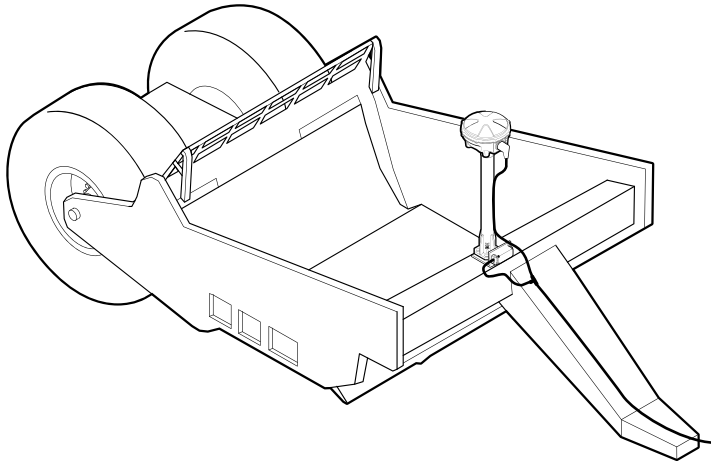
8.3 Cable routing

8.3.1 Main harness to cross tube

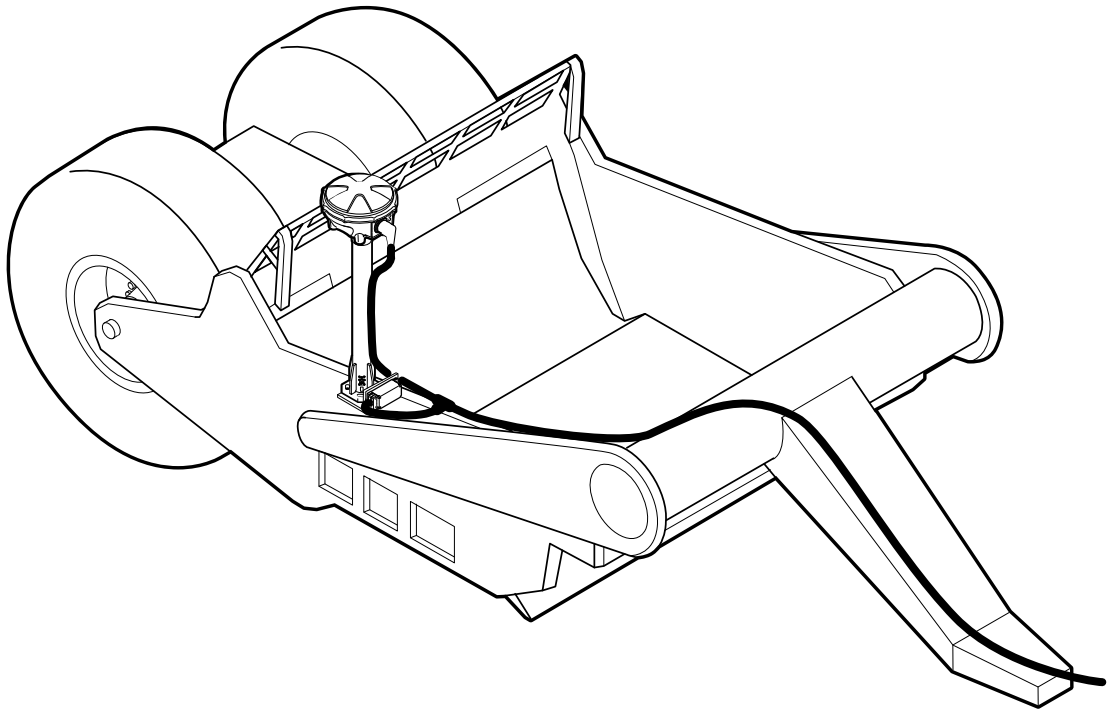
1. Route the main harness from where it comes out of the cab to the back of the machine to the quick disconnect housing. Use a harness extension as needed.
2. Route the harness from the quick disconnect housing along the hydraulic lines on the goose neck to the cross tube. Use the CL-Clamps to secure the cable at regular intervals.

3. Continue routing the harness to the mast mount.
4. Connect the EC520 platform harness, as shown in the connection diagram (96607-10).

The following illustration shows a possible routing solution for a system with the mast on the cross tube.



8.3.2 Main harness to the side of the bowl



8.4 Validate the GNSS system install

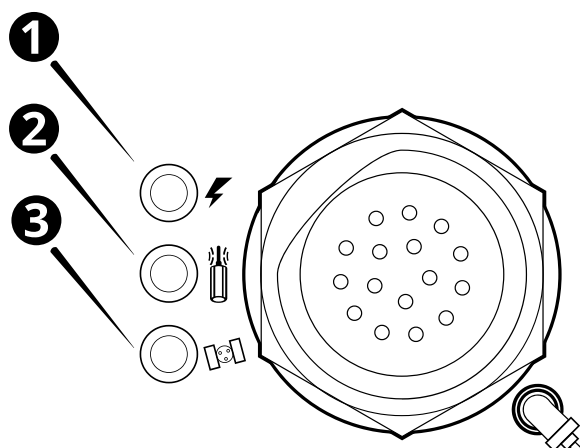
To validate the GNSS system installation:

- 8.4.1 Check GNSS receiver status indicators
- 8.4.2 Check radio power LED
- 8.4.3 Check device connections

8.4.1 Check GNSS receiver status indicators

The MS9xx GNSS receivers have three LEDs next to the harness connector. How these LEDs behave indicates the receiver status, as shown in the following figure.

In a dual GNSS system the left receiver data link should flash at 1 Hz and the right receiver should be solid.

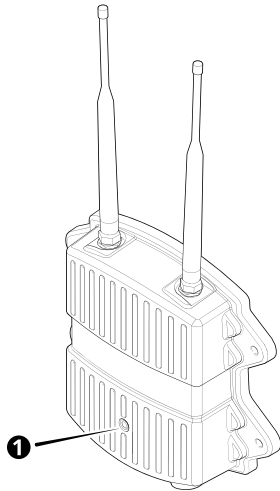


LED	Left	Right
❶ Power	Off – No power On – Power *	Off – No power On – Power *
❷ Data link	Off – No corrections 1 Hz – Corrections †	Off – No corrections On – Corrections received from the left receiver †
❸ Satellite	Off – Not tracking 2 Hz – Tracking fewer than 4 satellites 1 Hz – Tracking more than 4 satellites *	Off – Not tracking 2 Hz – Tracking fewer than 4 satellites 1 Hz – Tracking more than 4 satellites *

* required

† in a dual GNSS system the left receiver data link should flash at 1 Hz and the right receiver should be on (solid)

8.4.2 Check radio power LED



The radio is connected to power if the LED is permanently on or flashing.

8.4.3 Check device connections

In the Web Interface, navigate to Monitor > Onboard Devices, and check that the newly installed devices are shown as connected.

Supported third party tablet installation

In this chapter:

- ▶ [Overview of third-party tablet installation](#)
- ▶ [Third-party tablet installation](#)

9.1 Overview of third-party tablet installation

9.1.1 Purpose of this guide

The purpose of this guide is to enable the use of supported third-party tablets with the Earthworks system.

9.1.2 Supported third-party tablets

To run this release of the product on a third-party tablet, use a tablet that meets the requirements for Earthworks. Refer to the Earthworks release notes for more information.

9.1.3 Prerequisites

Before starting a third-party tablet installation:

- Make sure that the EC520 controller has the required OS and firmware installed, and that the controller's Wi-Fi access point is enabled. For more information, refer to *Earthworks Configure and Connect Guide*.
- It is strongly recommended that you review this entire guide. This will ensure you understand the full installation workflow.

9.1.4 Equipment list

NOTE –For users who supply their own display, the kit still includes a TD5x0 display harness, which must be used for connecting to the AA510.

- A PC
- Optionally, a cable to connect the tablet to a cigarette lighter for power

9.2 Third-party tablet installation

9.2.1 Download the app to the tablet

1. Turn on the tablet.
2. For Android 6 or 7 devices, navigate to Settings > Security > Unknown Source, and allow installation of apps from unknown sources.
3. Navigate to Settings > WLAN and connect to the EC520 controller:
 - The name of the access point is the SSID set during the configuration of the controller.
 - The password of the access point is the password set during the configuration of the controller.

NOTE – Some implementations of the Android operating system do not automatically reconnect to Wi-Fi access points that are not connected to the Internet. If a Network Error message displays after you turn the tablet off, and then turn it on again, check that Wi-Fi is enabled on the tablet, and that the controller is connected.

4. In a web browser, browse to: 192.168.168.1. The Web Interface log-in screen displays.

TIP – Earthworks supports the Google Chrome browser to run the Web Interface.

5. Log in to the Web Interface. The Web Interface Home screen displays.
6. Navigate to Advanced > Install Operator App. The Install Operator App screen displays.
7. Select Install Direct. The Earthworks firmware download begins.
8. On your device, locate the downloaded *launcher.apk* file, and select it.
9. Follow the instructions on the screen. For Android 8 and above, if the install is blocked by Android, set Allow from this source to ON.
10. If prompted, allow the plugin to access photos, media, and files on your device.

9.2.2 Locate and attach the tablet

In the cab:

1. Decide on how to attach the supported third-party tablet in the cab. We do not suggest locations or methods of attachment.
2. Optionally, attach the cable to connect the display to a cigarette lighter for power.

9.2.3 Validate the install

NOTE – To view the log-in screen, make sure at least one operator is configured, and that maintenance lockout is not active. For more information, refer to the *Configuration Guide*.

1. Turn on the tablet.
2. Locate and tap the app icon. If one of the following displays, then the tablet is connected to the EC520 controller:
 - the log-in screen
 - the *No operator configured for this machine* error message
 - the *Maintenance Lockout Active* error message

It is recommended that you install Android updates when available to keep the system up to date.

Torque charts

10.1 Imperial

Maximum torque values in FT-LB and Nm. Applicable to clean, dry, coarse threads in iron.

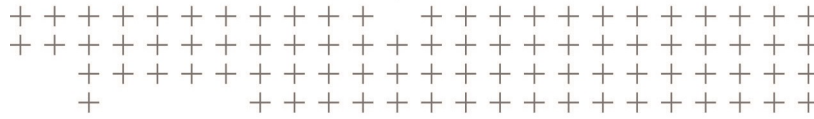
Nominal size (inches)											
	Grade	Torque	1/4	5/16	3/8	7/16	1/2	9/16	5/8	3/4	1"
	SAE 2	FT-LB	6	11	19	30	45	66	93		300
										150	
	SAE 5	FT-LB	9	18	31	50	75	110	150	250	580
		Nm	12	24	42	68	102	149	203	339	786
	SAE 8	FT-LB	13	29	46	75	115	165	225	370	890
		Nm	18	39	62	102	156	224	305	502	1207
	SOCKET HEAD	FT-LB	14	30	50	80	120	175	240	390	960
		Nm	19	41	68	108	163	237	325	529	1302

10.2 Metric

Maximum torque values in FT-LB and Nm. Applicable to clean, dry, coarse threads in iron.

Nominal size (mm)										
	Grade	Torque	6	8	10	12	14	16	18	20
	4.6, 4.8	FT-LB	3	8	16	27	43	60	90	120
		Nm	4	11	22	37	58	81	122	163
	8.8	FT-LB	7	17	33	59	100	130	180	280
		Nm	10	23	45	80	136	176	244	380

Nominal size (mm)										
	Grade	Torque	6	8	10	12	14	16	18	20
	10.9	FT-LB	10	25	48	85	130	200	280	400
		Nm	14	34	65	115	176	271	380	542
	12.9	FT-LB	12	29	57	100	150	230	–	–
		Nm	16	39	77	136	203	312	–	–



Notices

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